CampusConnect Project Comprehensive Report

Executive Summary: CampusConnect is a web application designed for university students to share course notes, collaborate on academic assignments, and organize study groups.

Architecture: Built with a decoupled client-server architecture. The frontend is built in React with TypeScript. The backend is built using Python FastAPI with asynchronous request handlers.

Database: PostgreSQL serves as the primary relational database, storing student profiles, course groups, and message threads.

Key Features: 1. Course-Specific Collaboration Channels. 2. Real-Time Peer-to-Peer Messaging. 3. Secure Academic File Uploads (PDF and DOCX). 4. Peer Rating and Content Moderation. 5. Role-Based Access Control.

Security: End-to-end token validation using JWT with hashed passwords and sanitized inputs.